Healthcare Patient & Hospital Performance Analysis

This project was created to analyze healthcare patient records and hospital performance data using Excel, SQL, Tableau, and Python. The main goal of the project was to understand patient information, medical conditions, billing patterns, age distribution, blood groups, and admission trends in hospitals.

Why This Project?

This project was selected because healthcare data analysis is one of the most important real-world applications of data analytics.
Hospitals generate huge amounts of patient data daily, and analyzing this data helps organizations make better decisions.

The project helps understand:
• Most common medical conditions
• Patient demographics
• Billing analysis
• Admission trends
• Healthcare patterns

This project also demonstrates practical skills required for Data Analyst roles.

My Contribution in the Project

In this project, I personally:
• Cleaned the dataset using Excel
• Identified and corrected missing values and formatting issues
• Imported and managed data in MySQL
• Wrote SQL queries for analysis
• Built Tableau dashboards for visualization
• Used Python and Pandas for data analysis and charts
• Analyzed healthcare insights from the dataset

Tools Used in the Project

1. Excel
Used for:
• Data cleaning
• Removing duplicates
• Finding blank values
• Formatting names and columns

2. MySQL
Used for:

• Database creation
• Query execution
• Data analysis using SQL

3. Tableau
Used for:
• Interactive dashboards
• Visual analytics
• Charts and graphs

4. Python
Used for:
• Data analysis
• Data visualization
• Exploratory Data Analysis (EDA)

Excel Work Done

Excel was used as the first step of data cleaning.

Important operations performed:
• COUNTBLANK() function to identify missing values
• Duplicate checking
• Proper formatting of names
• Data validation
• Data correction

Example Formula:
=COUNTBLANK(A:A)

This formula counts empty cells in a column.

SQL Implementation

The cleaned dataset was imported into MySQL Workbench.

Steps followed:
1. Created database
2. Imported CSV file
3. Queried data using SQL

Important SQL Queries Used:

1. Use Database
USE hospital_analysis;

Purpose:
Selects the database for operations.

2. Total Records
SELECT COUNT(*) FROM hospital_clean;

Purpose:
Counts total rows in the table.

3. Average Billing
```
SELECT AVG(CAST(REPLACE(`Billing Amount`, ',', '') AS DECIMAL(10,2))) AS average_bill
FROM hospital_clean;
```

Purpose:
Calculates average billing amount.

4. Medical Condition Analysis
```
SELECT `Medical Condition`, COUNT(*) AS total_cases
FROM hospital_clean
GROUP BY `Medical Condition`
ORDER BY total_cases DESC;
```

Purpose:
Finds most common diseases.

5. Duplicate Check
```
SELECT Name, COUNT(*)
FROM hospital_clean
GROUP BY Name
HAVING COUNT(*) > 1;
```

Purpose:
Identifies duplicate patient names.

Tableau Dashboard Implementation

Tableau was used to build visual dashboards.

Charts Created:
• Gender Distribution
• Blood Group Distribution
• Medical Condition Analysis
• Age Distribution Histogram
• Average Billing Analysis

Dashboard Features:
• Interactive charts
• Clean layout
• Multiple visual insights
• Healthcare trend analysis

Insights:
• Arthritis and Diabetes were among the most common conditions.
• Male and Female patient counts were almost equal.
• Average billing amount was around 25K.

Python Implementation

Python was used for data analysis and visualization.

Libraries Used:
• Pandas
• Matplotlib
• Seaborn
• Openpyxl

Important Python Concepts Used:
• DataFrame operations
• Data loading
• Missing value checking
• Chart generation

Main Python Code Explanation:

1. Import Libraries
```
import pandas as pd
```

Purpose:
Imports Pandas library for data analysis.

2. Load Dataset
```
df = pd.read_csv("hospital_cleaned.csv")
```

Purpose:
Loads CSV dataset into DataFrame.

3. Check Missing Values
```
df.isnull().sum()
```

Purpose:
Counts missing values in each column.

4. Gender Analysis
```
df['Gender'].value_counts()
```

Purpose:
Counts male and female patients.

5. Medical Condition Analysis
```
df['Medical Condition'].value_counts()
```

Purpose:
Counts disease occurrences.

6. Average Billing
```
df['Billing Amount'].mean()
```

Purpose:
Calculates average billing amount.

7. Visualization
```
sns.countplot(x='Gender', data=df)
```

Purpose:
Creates gender distribution chart.

Project Results

Final results from the project:
• Total Patients: 54,966
• No missing values after cleaning
• Average Billing Amount: ~25,544
• Most common medical conditions:
- Arthritis
- Diabetes
- Hypertension

The project successfully combined multiple technologies for complete healthcare analysis.

Possible Improvements

Future improvements that can be added:
• KPI cards in Tableau
• Interactive filters
• Monthly trend analysis
• Correlation heatmaps
• Machine Learning prediction models
• Hospital performance scoring
• Real-time dashboards
• Power BI integration

Interview Questions and Possible Changes

Interviewers may ask:

1. Why did you choose this project?
Answer:
Healthcare analytics is a real-world domain with large datasets and meaningful insights.

2. Why did you use Tableau?
Answer:
Tableau provides interactive and professional dashboards.

3. Why did you use Python?
Answer:
Python helps automate analysis and perform advanced visualization.

4. What challenges did you face?
Answer:
• CSV import issues
• SQL column formatting problems
• Tableau data connection issues
• Python setup and file path errors

5. What changes can be made?
Possible changes:

- Add filters and slicers
- Add predictive analytics
- Add machine learning
- Add hospital comparison dashboards
- Add time-based analysis
- Use APIs for live healthcare data

6. Why did you use SQL?
Answer:
SQL helps store, manage, and query large structured datasets efficiently.

7. What is the role of Excel?
Answer:
Excel was used mainly for initial cleaning and preprocessing.

Conclusion

This project demonstrates end-to-end data analysis using Excel, SQL, Tableau, and Python.
It helped build practical skills in data cleaning, querying, visualization, and analytical thinking.

The project is suitable for:
- Data Analyst interviews
- Internship applications
- Resume projects
- Portfolio demonstrations

Overall, the project shows the ability to work with real-world healthcare data and derive meaningful insights using multiple analytical tools.